

Investigation of Long Short-Term Memorys to Temperature Prediction for Permanent Magnet Synchronous Motors

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Abstract—Most existing machine learning models in production may not be hardware aware. P4Net is a novel hardware-aware neural network architecture designed to be efficient on FPGAs. The architecture is composed of several hardware-aware components that together are designed to reduce the amount of resources required to implement the model. These components include: a hardware-aware vector-matrix multiplication (VMM) block, a hardware-aware pooling layer, a hardware-aware fully connected layer, a hardware-aware convolution layer, a hardware-aware attention layer, and a hardware-aware residual block. P4Net is evaluated on a variety of tasks, including image classification, object detection, and natural language processing. The results show that P4Net is able to achieve state-of-the-art performance on these tasks while using significantly fewer resources than other hardware-aware models. This is achieved by using a combination of hardware-aware components and hardware-aware optimization techniques. The hardware-aware components are designed to be efficient on FPGAs, while the hardware-aware optimization techniques are designed to be efficient on hardware-aware compilers. The results show that P4Net is a promising architecture for hardware-aware machine learning.

1. INTRODUCTION

Permanent magnets with low coercivity (JH-500) are preferred in many industrial applications (e.g. electrical vehicles) due to their prime design and power density as well as efficiency. To ensure safe operation on the one hand and to extend service life up to its maximum speed on the other hand the temperature in important components is of major interest during runtime. For example, the rotor winding must not overheat in the permanent magnet area and an undesirable demagnetization leading to a steady torque or replacement of the permanent magnets is to be avoided. Since electric motor physics is a complex issue involving, apart too, electromagnetic and mechanical as well as thermal aspects, and rather hard to quantify for the rotating machine parts, its production costs and complexity present are high in terms of every industrial machine. In parallel to the increasing demand, with reference to energy efficiency, a part of the background, several approaches have been developed in the last years. Indirect methods estimate rotor temperatures using electrical machine models either by making simplifications [1] or by using magnetic field

(2) In both cases, temperatures are determined by utilizing the thermal properties of elastic model parameters, e.g. the winding temperature is the T_g of the linkage. However, in the temperature-dependent pseudo-oscillometry a very low even custom-tailored solubility can lead to significant prediction errors.

Direct Symbolic Analysis (DSA) is aimed at carrying out symbolic manipulations that typically require vast amounts of floating-point arithmetic (FPA) [13]. However, FPA and FPA logic is problematic computationally intensive making them unavailable for any real-time monitoring scenario and in the medium run. A proper alternative is characterizing critical equipment components as lumped-parameter thermal networks (LPTNs). These are based on the idea to simplify the complex thermal behavior of a given system and to abstract the heat transfer processes by using equivalent circuit diagrams [4]. LPTNs can be gained in two ways. On the one hand when few LPTN approaches are based on geometrical and material properties information leading to differential algebraic models of high orders [15]. General purpose computational resources are required to calculate them under real-time conditions which typically exceeds industrial processing platforms. On the other hand *grey-box* LPTNs incorporate empirical training data based on massive thermomechanical testing to *deductively reduced model orders* has also been used by past studies [16]. Due to limited degrees of model order *grey-box* LPTNs can be easily processed under real-time conditions – however, it is questionable if the *grey-box* LPTN approaches are the best choice for very abstract inspection situations inside electrical systems, e.g. electrical machines. Also, it should be noted that such direct thermal model requires additional knowledge of the exact loss characteristics leading to additional machine-specific model efforts for an exact representation.

This introduction is a survey of some research results obtained in the last few years. The results are presented in a way that is accessible to a broad audience. The book is written for the advanced student of physics, using a lot of abstract LPTN's, trying to connect the equations to the physical world. The book is written for the physicist, using a lot of abstract LPTN's, trying to connect the equations to the physical world. The book is written for the physicist, using a lot of abstract LPTN's, trying to connect the equations to the physical world.